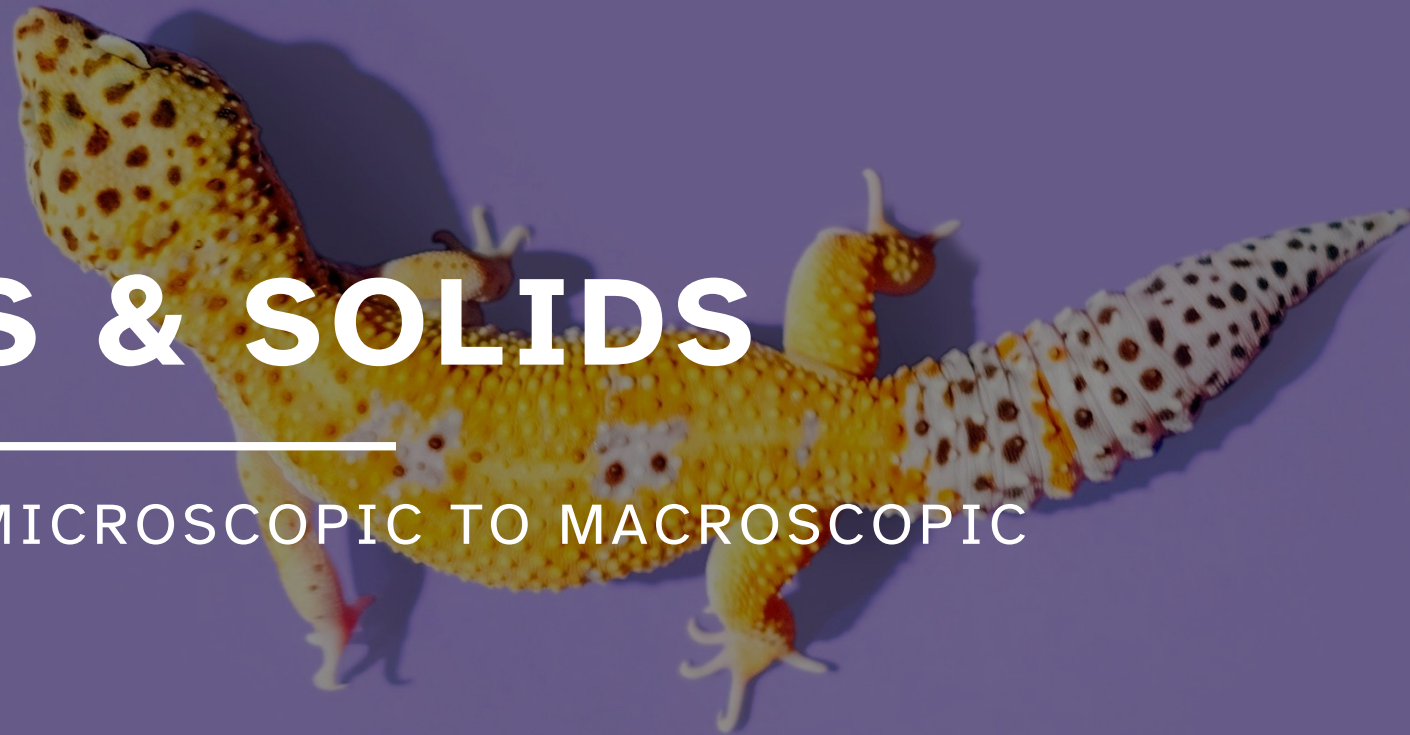


12C LIQUIDS & SOLIDS

A leopard gecko with yellow and brown spots is positioned in the center of the slide, facing left. The gecko's body is covered in numerous small, dark brown spots on a lighter yellow background. Its tail is long and tapers, also featuring a pattern of spots. The gecko is set against a solid purple background.

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 12

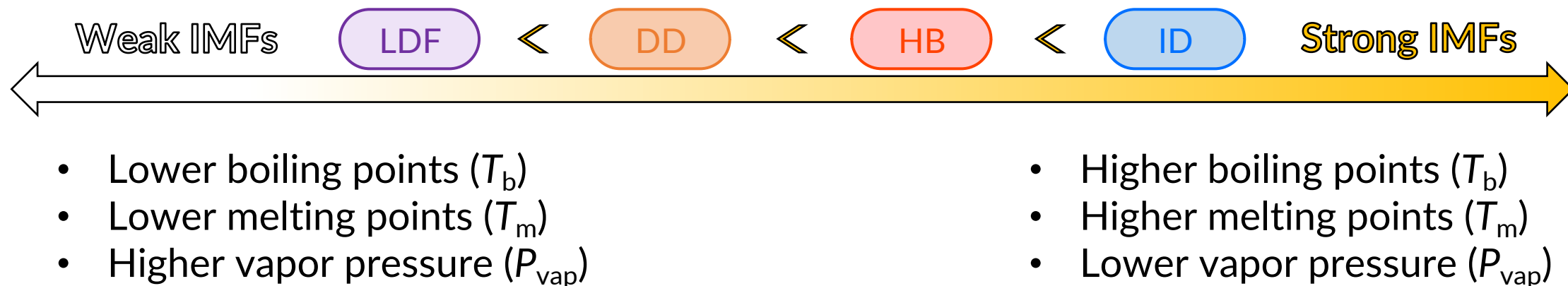
1. List and describe the four types of intermolecular forces and identify the predominant intermolecular force in a substance.
2. Define and apply terms related to phase changes.
3. Calculate energy changes related to phase changes.
4. Interpret heating curves.
5. Predict physical properties (relative vapor pressures and relative phase change temperatures) based on strengths of intermolecular forces.
6. Define, describe, and interpret phase diagrams.
7. Describe the structures of various cubic unit cells.

Intermolecular Forces

12.1

Physical molecular properties can be predicted from the types and strengths of IMFs.

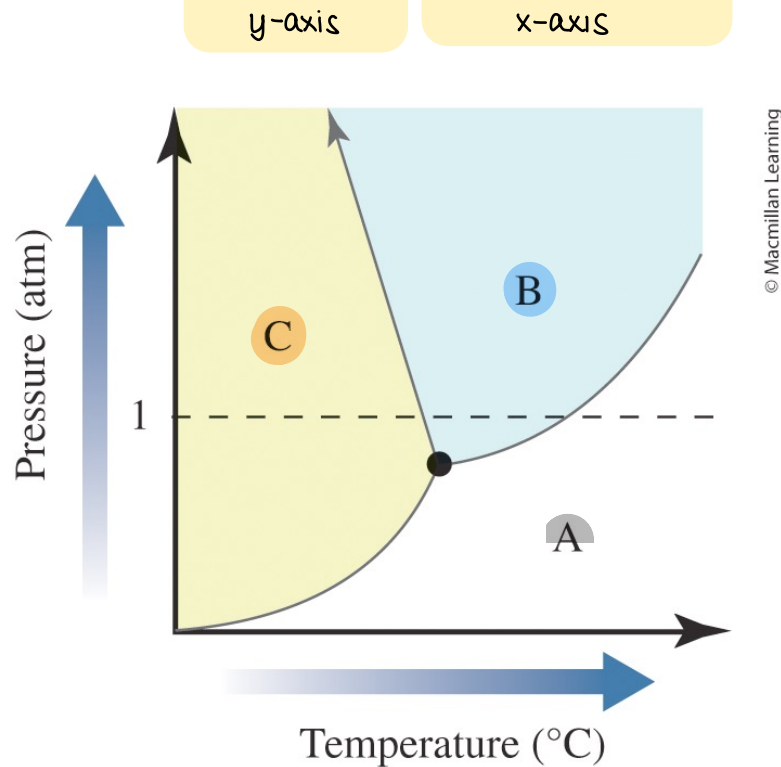
Only IMFs are broken during phase changes (no chemical bonds are broken).



Phase Diagrams

12.5

A phase diagram shows the most energetically stable phases for a substance under all possible pressure–temperature (P, T) combinations.



- **Solid Phase:** high pressures, low temperatures → C
- **Liquid Phase:** moderate pressures and temperatures → B
- **Gas Phase:** low pressures, high temperatures → A
- **Co-existence or boundary lines:** All the (P, T) conditions at which two phases co-exist in equilibrium.
these are transition temperatures!

Phase Diagrams

12.5

A phase diagram shows the most energetically stable phases for a substance under all possible pressure–temperature (P, T) combinations.

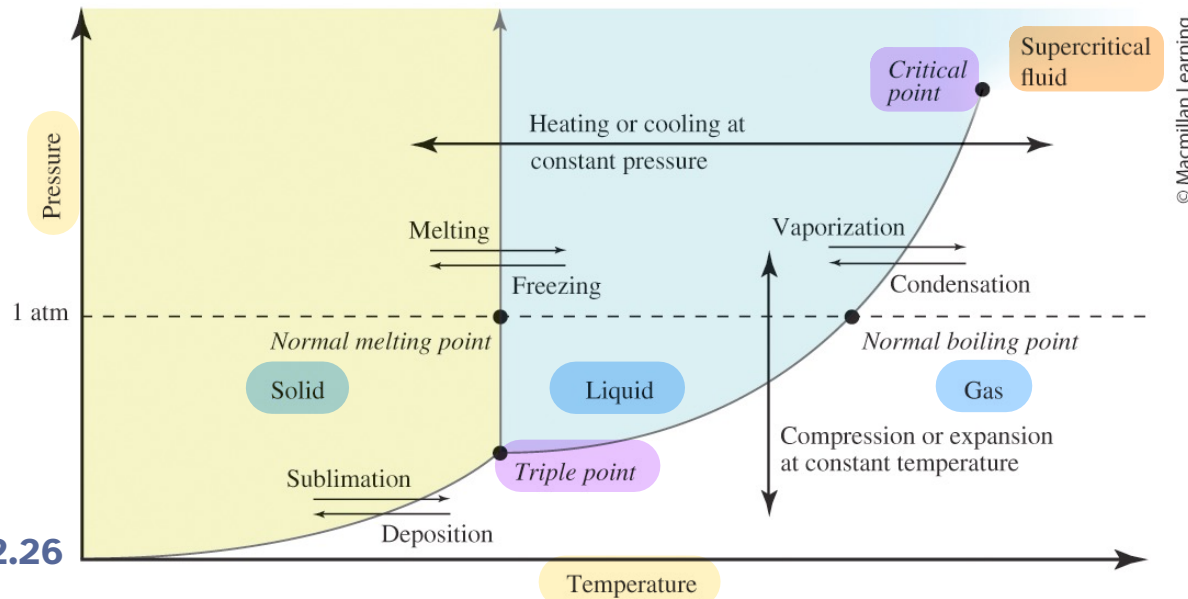


FIGURE 12.26

phases highlighted

lines separating phases
are P,T conditions
where phases co-exist

- Triple point: single point at which three phase boundary lines intersect. At this (P, T) all three phases co-exist.
- Critical point: The (P, T) above which the substance exists as neither liquid nor gas, but as a **supercritical fluid**, which is a fourth phase with properties common to both liquids and gases.

IN-CLASS QUESTION 1

Learning Objective(s): 12.6

The phase diagram for CO_2 is drawn to the right.

for each statement relevant P,T marked as ○ ● ⊕ ⊗ ⊕

Which of the following statements are **true**?

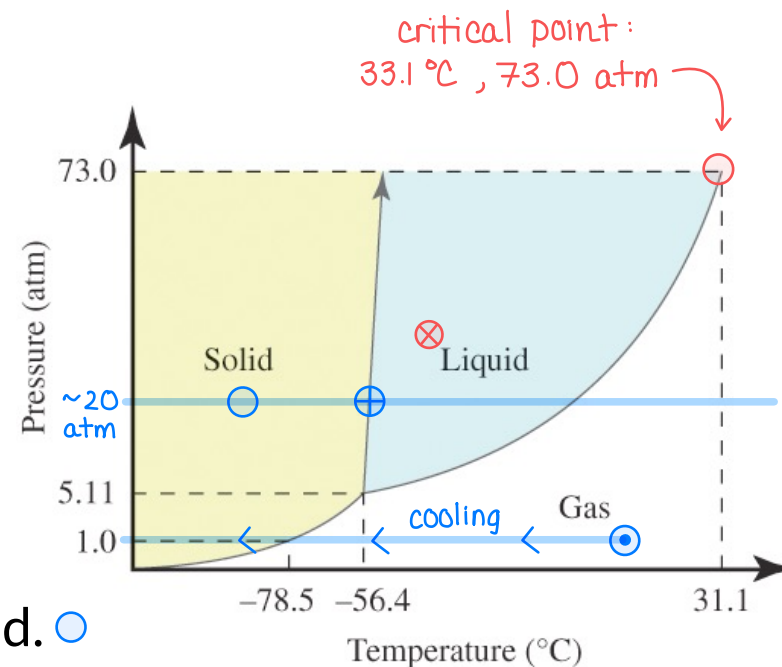
☒ The critical point occurs at ~~-56.4°C~~ and ~~5.11 atm~~ . ○

☒ The most stable phase at 20.0 atm and -95°C is the solid. ○

☒ If the gas is cooled from 25°C at 1.00 atm , then it undergoes deposition. ●
gas → solid

☒ The ~~supercritical fluid~~ is the most stable phase at 30.0 atm and -45°C . ⊗ = liquid

☒ The melting point at 20.0 atm is about -50°C . ⊕

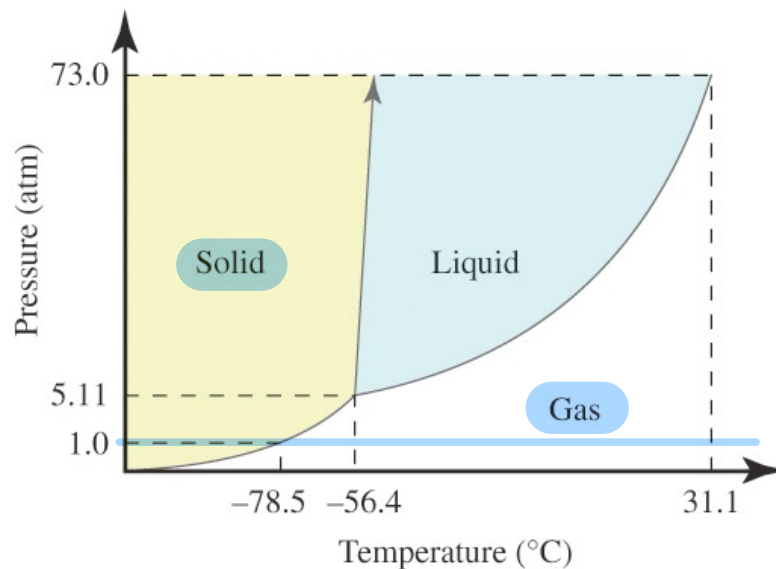


EXAMPLE PROBLEM

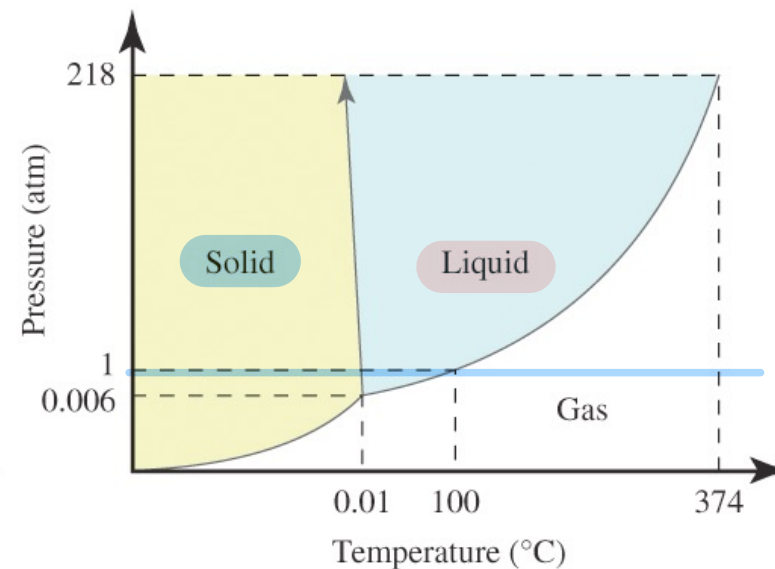
Learning Objective(s): 12.6

Considering the phase diagrams for water and carbon dioxide shown below, which substance can **sublime at 1 atm**?

solid $\rightarrow$ gas



(a) Carbon dioxide ✓



(b) Water ✗

Phase Diagrams

12.5

Crystalline solids are solids whose constituent particles (atoms, ions, or molecules) have an *ordered* arrangement extending over a long range. There are a few different types:

- Molecular solids: $\text{H}_2\text{O}(\text{s})$ and $\text{I}_2(\text{s})$
- Ionic solids: $\text{NaCl}(\text{s})$ and $\text{ZnS}(\text{s})$
- Covalent network solids: $\text{C}(\text{s})$, $\text{Si}(\text{s})$, and $\text{SiO}_2(\text{s})$
- Metallic solids

We will only focus on metallic solids!

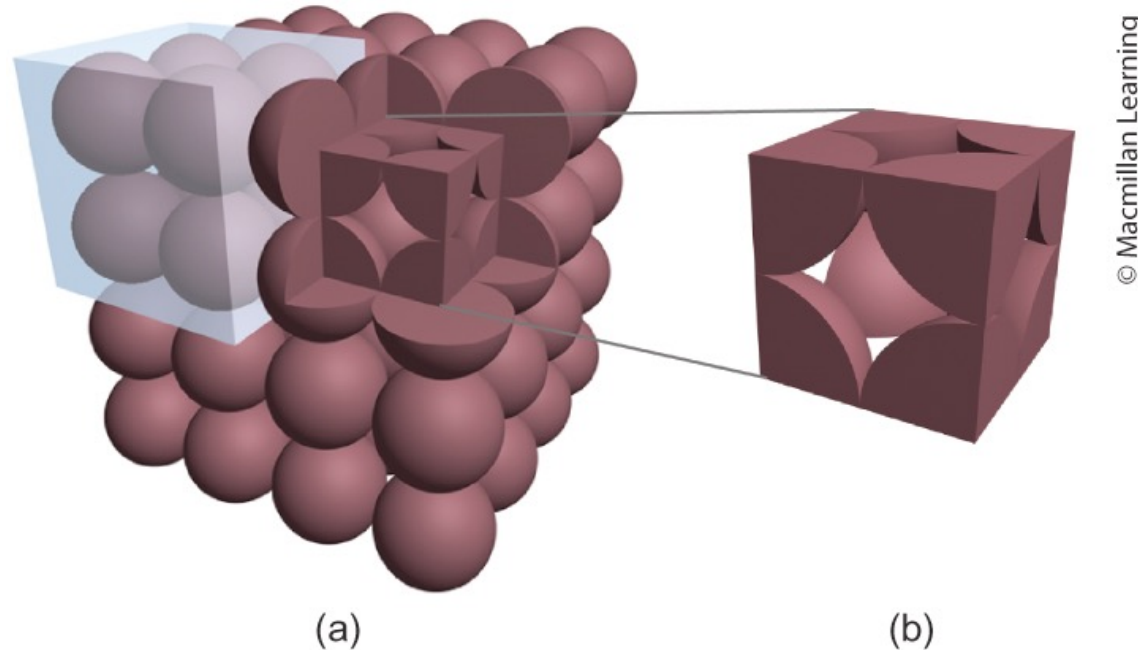
Note: If the constituent particles are *randomly* arranged and have no ordered long-range structure, we call them *amorphous solids*.

The Unit Cell and the Structure of Crystalline Solids

12.7

Metallic solids are solids whose constituent particles are neutral, monatomic metal atoms in a structured 3D array.

- The unit cell is the simplest repeating unit of the crystal structure and arises from how the layers of atoms are arranged.



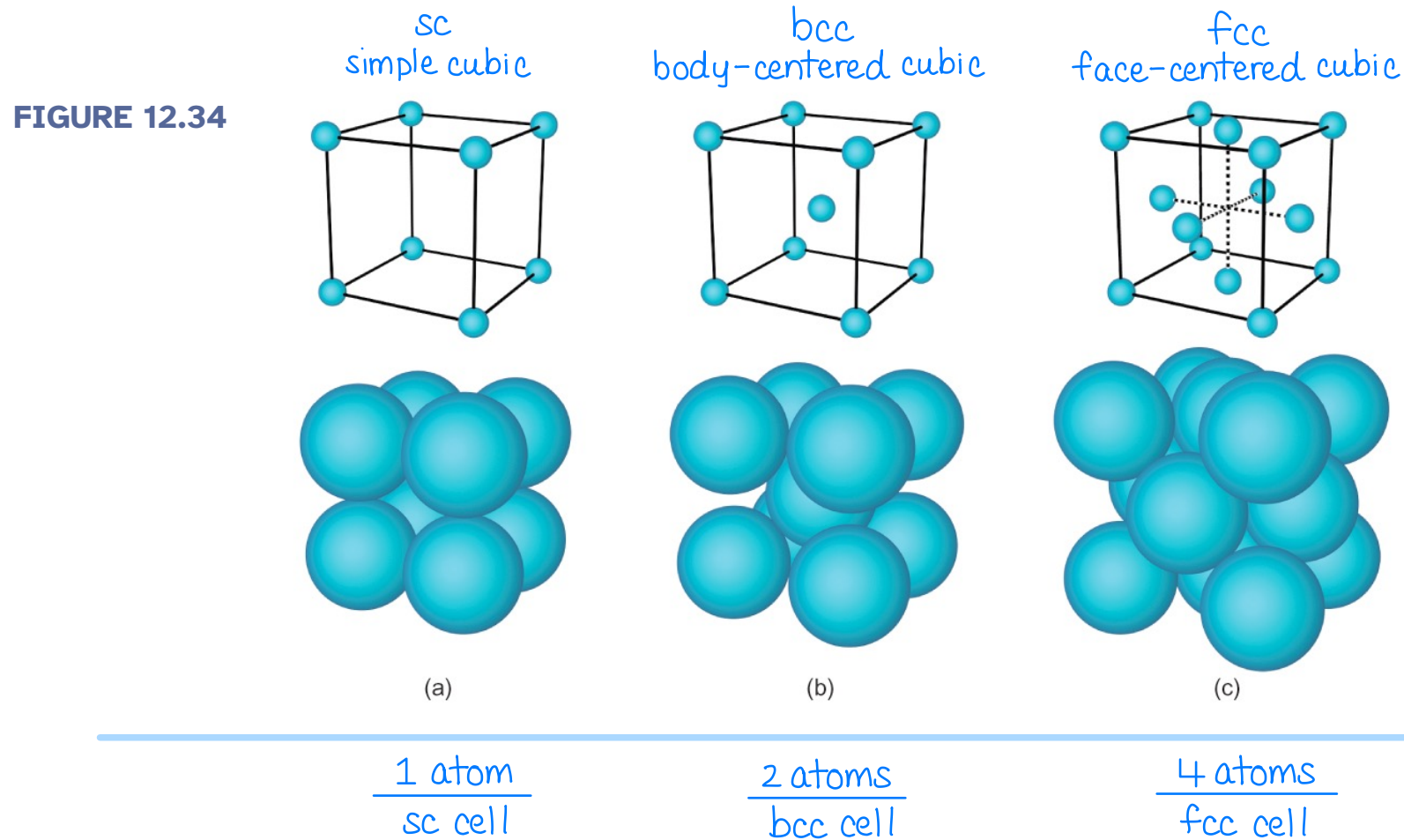
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FIGURE 12.33

The Unit Cell and the Structure of Crystalline Solids

12.7

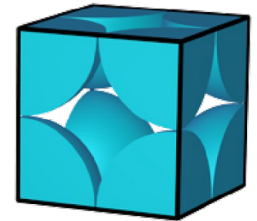
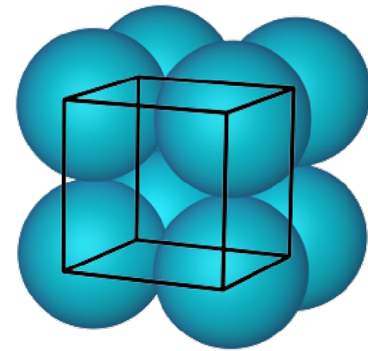
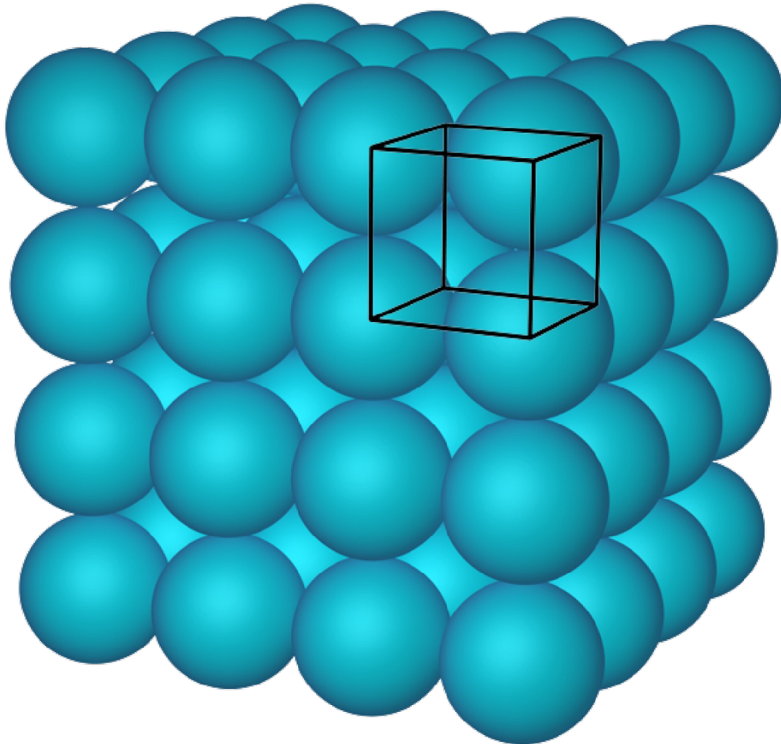
There are *only* 3 unit cell structures we will consider.



The Unit Cell and the Structure of Crystalline Solids

12.7

Counting atoms in a simple cubic unit cell.



8 vertices in a cube
at each vertex: $\frac{1}{8}$ atom inside cube
 $8 \times \frac{1}{8} \text{ atoms} = 1 \text{ atom per sc unit cell}$

FIGURE 12.35

The Unit Cell and the Structure of Crystalline Solids

12.7

Counting atoms in a body-centered cubic unit cell.

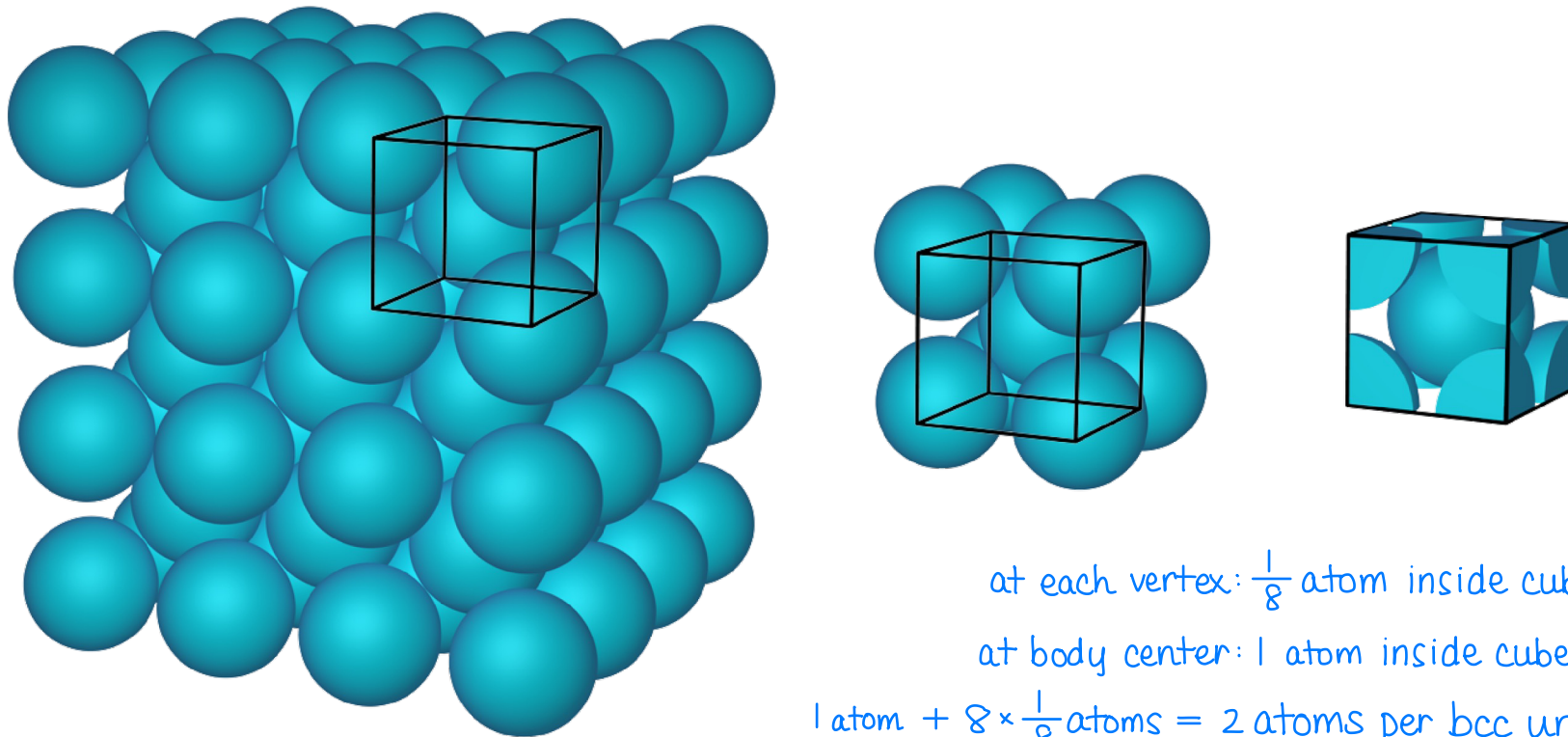


FIGURE 12.36

The Unit Cell and the Structure of Crystalline Solids

12.7

Counting atoms in a face-centered cubic unit cell.

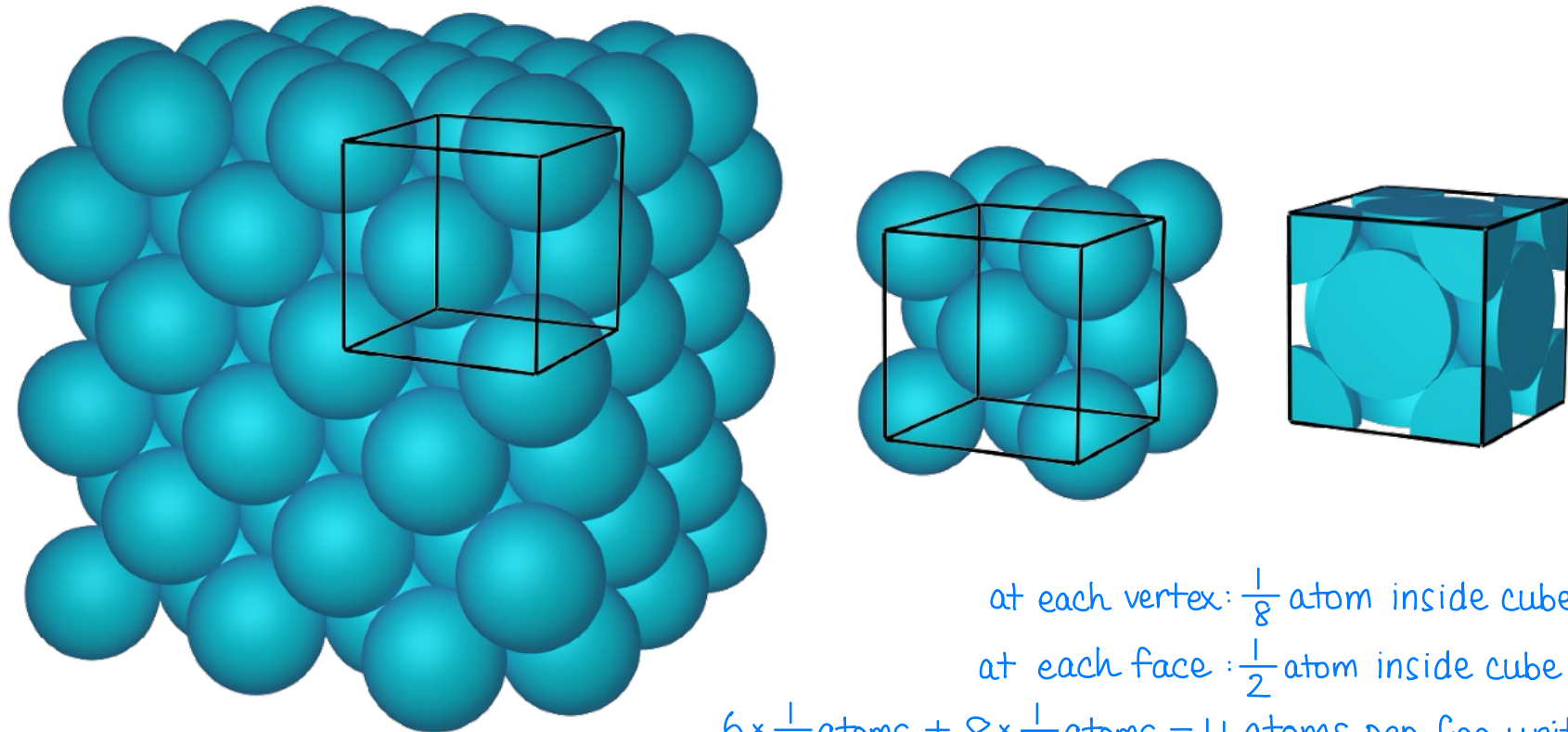


FIGURE 12.37

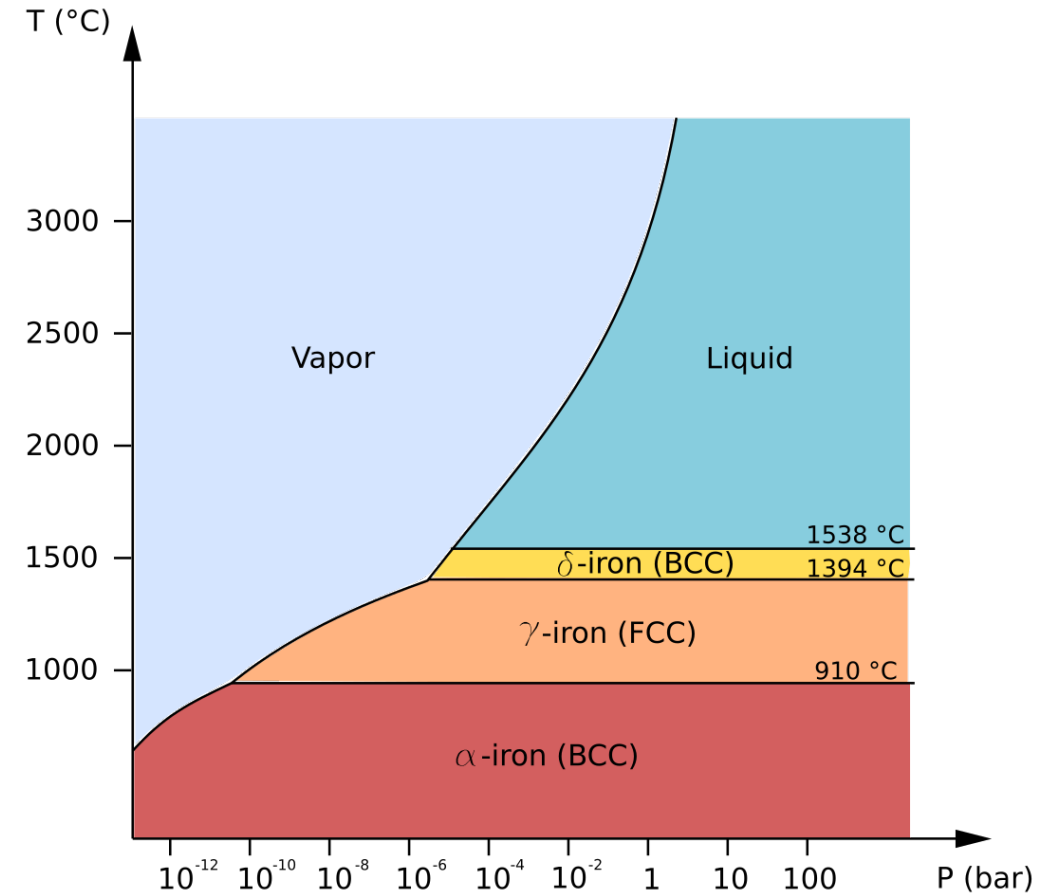
IN-CLASS QUESTION 2

Learning Objective(s): 12.7

Upon heating iron transitions from *bcc* (α -iron) to *fcc* (γ -iron) packing.

Which is expected to have the higher density?

- ☐ *bcc* (α -iron) 2 atoms
bcc cell
- ☒ *fcc* (γ -iron) 4 atoms
fcc cell



BONUS QUESTION

Assessed on: Fall 2021, Exam 3

If you *decrease* the temperature from 9 °C to 3 °C at a constant pressure of 58 torr, the substance undergoes which process?

- ☐ Freezing
- ☐ Melting
- ☐ Condensation
- ☐ Vaporization
- ☐ Sublimation
- ☐ Deposition

